Bachelor's Thesis Computing Science



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Thesis Title

Subtitle if you like

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Abstract

Brief outline of research questions, results. (The preferred size of an abstract is one paragraph or one page of text.)

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Chapter 1

Introduction

The introduction of your bachelor thesis introduces the research area, the research hypothesis, and the scientific contributions of your work. A good narrative structure is the one suggested by Simon Peyton Jones [1]:

- describe the problem / research question
- motivate why this problem must be solved
- demonstrate that a (new) solution is needed
- explain the intuition behind your solution
- motivate why / how your solution solves the problem (this is technical)
- explain how it compares with related work

Close the introduction with a paragraph in which the content of the next chapters is briefly mentioned (one sentence per chapter).

Starting a new paragraph is done by inserting an empty line like this.

Chapter 2

Lambda Calculus

This section provides a detailed and formal description of the λ-calculus. We define a formal grammar of the λ-calculus and give examples of some terms in the λ-calculus. Then we explain β-reduction, after which we define call-by-name evaluation and call-by-value evaluation.

2.1 Introduction to the λ-calculus

The grammar of the lambda calculus is as follows:

Definition 2.1 (λ-calculus)**.** The terms of the λ-calculus are given by:

$$M, N, P, Q \quad ::= \quad x \mid \lambda x.M \mid MN$$

Terms are denoted by M, N, P or Q and can either be of the form x, $\lambda x.M$ or MN:

- x is a variable, which is a symbol that represents an input.
- $\lambda x.M$ is an abstraction. An abstraction is an anonymous function, where x is the parameter and M is the body of the function.
- MN is a function application, where M and N are terms.

The λ-term $\lambda x.x$ is an abstraction. This specific abstraction is called the identity function and has one input parameter, namely x, and returns the input x. The body of the function is also x in this case and the function is often abbreviated as **I**. The λ-term $(\lambda x.x)y$ is a function application. So the identity function is applied to the variable y and the λ-term reduces to the variable y.

2.2 β-Reduction

Before we discuss β-reduction, we first define bound and free variables and substitution.

2.2.1 Variables

Definition 2.2 (Free Variables)**.** Let us first define free variables. Let P be any term in the λ-calculus and $FV(P)$ be the set that contains all free variables in P. We define FV inductively on P. Since P can only be a variable, an abstraction or a function application, P can only be of the form x, $\lambda x.M$ and MN. Therefore, we define FV as follows:

$$\begin{aligned} FV(x) &= \{x\} \\ FV(\lambda x.M) &= FV(M) \backslash \{x\} \\ FV(MN) &= FV(M) \cup FV(N) \end{aligned}$$

The variable in a λ-term that solely consists of a variable is free. All free variables in an abstraction are those that are not the parameter of the abstraction. For example, the λ-term $\lambda x.yx$ has one free variable, y. We can use the following reasoning:

Example 2.3. Let us now reason that $FV(\lambda x.yx) = \{y\}$.

$$\begin{aligned} FV(\lambda x.yx) &= FV(yx) \backslash \{x\} \\ &= (FV(x) \cup FV(y)) \backslash \{x\} \\ &= (\{x\} \cup \{y\}) \backslash \{x\} \\ &= \{x, y\} \backslash \{x\} \\ &= \{y\} \end{aligned}$$

Definition 2.4 (Bound Variables)**.** Let us now define bound variables. Again, let P be any term in the λ-calculus and $BV(P)$ be the set of all bound variables in P. We define BV inductively on P as follows:

$$\begin{aligned} BV(x) &= \emptyset \\ BV(\lambda x.M) &= BV(M) \cup \{x\} \\ BV(MN) &= BV(M) \cup BV(N) \end{aligned}$$

We say that the abstraction $\lambda x.M$ binds the variable x.

Example 2.5. Using our previous example of the λ-term $\lambda x.yx$, we can now reason that $BV(\lambda x.xy) = \{x\}$.

$$\begin{aligned} BV(\lambda x.xy) &= BV(xy) \cup \{x\} \\ &= (BV(x) \cup BV(y)) \cup \{x\} \\ &= (\emptyset \cup \emptyset) \cup \{x\} \\ &= \{x\} \end{aligned}$$

The following example will demonstrate that using two different variables with the same name in a λ-term is quite tricky: $(\lambda x.x)x$. By the definitions

of FV and BV, we can reason that $FV((\lambda x.x)x) = \{x\}$ and $BV((\lambda x.x)x) = \{x\}$. This is quite confusing and we prefer to write $(\lambda y.y)x$ instead, where y is a new variable with a different name. We now have $FV((\lambda y.y)x) = \{x\}$ and $BV((\lambda y.y)x) = \{y\}$. These λ-terms are equivalent by the notion of *alpha equivalence*, which states that λ-terms that differ only by the names of bound variables are considered the same. We will discuss this in more detail after we define substitution.

2.2.2 Substitution

Keeping the definitions of free and bound variables in mind, we now define substitution.

Definition 2.6 (Substitution)**.** In this research, substitution is often denoted as $P[x \mapsto Q]$ and defined inductively on P by:

$$\begin{aligned}
x[x \mapsto Q] &= Q \\
y[x \mapsto Q] &= y \text{ if } (x \neq y) \\
(MN)[x \mapsto Q] &= M[x \mapsto Q]N[x \mapsto Q] \\
(\lambda x.M)[x \mapsto Q] &= \lambda x.M \\
(\lambda y.M)[x \mapsto Q] &= \lambda z.M[y \mapsto z][x \mapsto Q] \text{ if } (x \neq y)
\end{aligned}$$

with z a variable defined by:

1. If $x \notin FV(N)$ or $y \notin FV(M)$ then $z = y$
2. Otherwise, z can be any variable such that $z \notin FV(N)$ or $z \notin FV(M)$

The variable 'z' always exists. Choosing a fresh variable for z will always satisfy condition two. It might not be clear why need to use the variable z and one can expect that the following rule is good enough:

Definition 2.7. This is a simpler version of the substitution rule which is not correct:

$$(\lambda y.M)[x \mapsto Q] = \lambda z.M[x \mapsto Q] \text{ if } (x \neq y)$$

The following example will illustrate why the more complicated rule is necessary. Consider the substitution $(\lambda y.xy)[x \mapsto y]$. The free variable of λ-term $\lambda y.xy$ is x and the bound variable is y. If we use the simple rule above, we get that $(\lambda y.xy)[x \mapsto y] = \lambda y.yy$, the meaning of the λ-term changes. However, the free variable we substituted, is not free anymore. The variable y is not a free variable in the term that we substituted y in. Therefore, we change the bound variable y in $\lambda y.xy$ to a new variable z. We can pick z as z is not a free variable in xy. Following the more complex rule, we get:

$$\begin{aligned}
(\lambda y.xy)[x \mapsto y] &= \lambda z.(xy)[y \mapsto z][x \mapsto y] \\
&= \lambda z.yz
\end{aligned}$$

2.2.3 α equivalence

Here, we define α equivalence. The idea is that renaming bound variables does not change the meaning of a λ-term. For example, the λ-terms $\lambda x.x$ and $\lambda y.y$ are equivalent ($\lambda x.x \equiv_a \lambda y.y$), since renaming all bound variables can result in identical λ-terms. The λ-terms $\lambda x.x$ and $\lambda y.y$ represent the same idea. We can define the relation $P \equiv_a P'$ inductively on P and P' as follows:

Definition 2.8 (α equivalence)**.** Let x and y be variables. The relation $\equiv_a$ is defined as follows:

$$\begin{array}{ll} x \equiv_a y & \text{if } x = y \\ MN \equiv_a M'N' & \text{if } M \equiv_a M' \text{ and } N \equiv_a N' \\ \lambda x.M \equiv_a \lambda y.M' & \text{if } M \equiv_a M'[y \mapsto x] \text{ with } x = y \text{ or } x \notin FV(M') \end{array}$$

The last rule states that abstractions $\lambda x.M$ and $\lambda y.M'$ are equivalent if M and M' are equivalent after substituting y for x in M'. However, x should be equal to y or x should not be a free variable in M'.

Example 2.9. Consider the following:

- $\lambda x.zx$ is equivalent to $\lambda y.zy$, since $zy[y \mapsto x] = zx \equiv_a zx$ and x is not a free variable in zy.
- $\lambda z.zx$ is not to $\lambda y.zy$, since z is a free variable in zy and x is not equal to y.

2.2.4 β-Reduction

Now, we are ready to discuss β-reduction. In the λ-calculus, there is one way to simplify terms, which is β-reduction. We can define:

Definition 2.10 (β-reduction)**.** Let M and N be terms in the λ-calculus. The relation $\to$ is defined as follows:

$$(\lambda x.M)N \to M[x \mapsto N] \qquad (\beta)$$

So a λ-term of the form $(\lambda x.M)N$ can be reduced by substituting x by N in M. However, without additional rules, we cannot apply β-reduction to subterms. For example, the following reduction would not be possible: $\lambda x.(\lambda y.y)x \to \lambda x.x$. Therefore, we can define the following rules:

Definition 2.11 (Closure rules)**.** Closure rules μ, ν and ξ are defined as follows:

$$\frac{M \to M'}{MN \to M'N}\,(\mu) \qquad \frac{N \to N'}{MN \to MN'}\,(\nu) \qquad \frac{M \to M'}{\lambda x.M \to \lambda x.M'}\,(\xi)$$

Now, we define $\to_\beta$ as β closed under μ, ν and ξ. So with $\to_\beta$ we can use β-reduction on all subterms if the subterm is of the form $(\lambda x.M)N$. A subterm of the form $(\lambda x.M)N$ is called a β-redex. In one reduction, we may need to use multiple rules. The β-redex that is reduced is underlined. For instance, we need to use the μ and ξ rule in the last example of the following reductions:

Example 2.12. This example shows how to use the closure rules.

$$\begin{aligned}
(\underline{\mathbf{II}})(\mathbf{II}) &\to_\beta \mathbf{I}(\mathbf{II}) && \beta \text{ with } \mu \text{ rule}\\
(\mathbf{II})(\underline{\mathbf{II}}) &\to_\beta (\mathbf{II})\mathbf{I} && \beta \text{ with } \nu \text{ rule}\\
\lambda x.\underline{\mathbf{I}x} &\to_\beta \lambda x.x && \beta \text{ with } \xi \text{ rule}\\
(\mathbf{I}(\lambda x.\underline{\mathbf{I}x}))(\mathbf{II}) &\to_\beta (\mathbf{I}(\lambda x.x))(\mathbf{II}) && \beta \text{ with } \mu \text{ and } \xi \text{ rule}
\end{aligned}$$

2.3 Call-by-name calculus

In this section, we discuss the call-by-name λ-calculus. The reduction rule that is used in this calculus is the same as the β reduction rule. However, we name the rule β_n to make it clear that it is the reduction rule of the call-by-name λ-calculus.

Definition 2.13 (Call-by-name reduction)**.** Let M and N be terms in the λ-calculus. The relation $\to_n$ is defined as follows:

$$(\lambda x.M)N \to M[x \mapsto N] \qquad \beta_n$$

We can define weak reduction, $\to_w$, as β_n closed under μ and ν. So $\to_w$ has the only restiction that it cannot reduce under λ's. The relation $\to_n$ is defined as β_n closed under μ. So using name evaluation, we can not evaluate the argument of a function. Call-by-name evaluation is defined as $\to_n^*$. In the following example, we give the call-by-name evaluation of the λ-term $(\mathbf{I}(\lambda x.\mathbf{I}x))(\mathbf{II})$. The β-redex that is reduced is underlined. Note that $\to_n$ is deterministic, so there is only one way to apply $\to_n$ on a λ-term.

Example 2.14. This is an example of call-by-name evaluation in λ_n.

$$\begin{aligned}
(\underline{\mathbf{I}(\lambda x.\mathbf{I}x)})(\mathbf{II}) &\to_n \underline{(\lambda x.\mathbf{I}x)(\mathbf{II})}\\
&\to_n \underline{\mathbf{I}(\mathbf{II})}\\
&\to_n \underline{\mathbf{II}}\\
&\to_n \mathbf{I}
\end{aligned}$$

Relation	$\boldsymbol{\beta}$	$\boldsymbol{\mu}$	$\boldsymbol{\nu}$	$\boldsymbol{\xi}$
$\rightarrow_{\beta_n}$	β_n	✓	✓	✓
$\rightarrow_n$	β_n	✓		
$\rightarrow_w$	β_n	✓	✓	

Table 2.1: Closure Rules for Different Reduction Relations

2.4 Call-by-value calculus

Before we can define the call-by-value λ-calculus (abbreviated as), we first define what we mean by values. Values are all λ-terms that are either a variable or an abstraction. Values are usually denoted by V or W.

Definition 2.15 (Values)**.** Values V are defined as follows:

$$V \quad ::= \quad x \mid \ \lambda x.M$$

In the λ_v, we can only reduce if the argument is a value. Therefore, the reduction rule is as follows:

Definition 2.16 (Call-by-value reduction)**.** Let M and V be terms in the λ-calculus with V a value. The relation $\rightarrow_v$ is defined as follows:

$$(\lambda x.M)V \rightarrow M[x \mapsto V] \qquad \beta_v$$

In order to define the $\rightarrow_v$ relation, we first define the rule $\nu_<$.

Definition 2.17 ($\nu_<$)**.** Let M and V be terms in the λ-calculus with V a value. The relation $\nu_<$ is defined as follows:

$$\frac{N \rightarrow N'}{VN \rightarrow VN'}\,\nu_<$$

The relation $\rightarrow_v$ can be defined as β_v, closed under μ and $\nu_<$. The $\nu_<$ forces evaluation from left to right and β_v makes sure that the argument needs to be a value. In the following example, we give the call-by-value evaluation of the λ-term $(\mathbf{I}(\lambda x.\mathbf{I}x))(\mathbf{II})$. The β-redex that is reduced is underlined. Note that $\rightarrow_v$ is deterministic, so there is only one way to apply $\rightarrow_v$ on a λ-term.

Example 2.18. This is an example of call-by-value evaluation in λ_n.

$$\begin{aligned}
(\underline{\mathbf{I}(\lambda x.\mathbf{I}x)})(\mathbf{II}) &\rightarrow_v (\lambda x.\mathbf{I}x)(\underline{\mathbf{II}}) \\
&\rightarrow_v \underline{(\lambda x.\mathbf{I}x)\mathbf{I}} \\
&\rightarrow_v \underline{\mathbf{II}} \\
&\rightarrow_v \mathbf{I}
\end{aligned}$$

Relation	$\boldsymbol{\beta}$	$\boldsymbol{\mu}$	$\boldsymbol{\nu}$	$\boldsymbol{\nu_<}$	$\boldsymbol{\xi}$
$\rightarrow_{\beta_n}$	β_n	✓	✓	✓	✓
$\rightarrow_n$	β_n	✓			
$\rightarrow_w$	β_n	✓	✓		
$\rightarrow_{\beta_v}$	β_v	✓	✓	✓	✓
$\rightarrow_v$	β_v	✓		✓	

Table 2.2: Closure Rules for Different Reduction Relations

Chapter 3

Rules

3.1 Rules Intuitionistic Logic

$$\frac{}{A \vdash A}\ \text{Id} \qquad \frac{\Gamma, \Delta \vdash A}{\Delta, \Gamma \vdash A}\ \text{Exchange}$$

$$\frac{\Gamma, A, A \vdash B}{\Gamma, A \vdash B}\ \text{Contraction} \qquad \frac{\Gamma \vdash B}{\Gamma, A \vdash B}\ \text{Weakening}$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B}\ \to\text{-I} \qquad \frac{\Gamma \vdash A \to B \qquad \Delta \vdash A}{\Gamma, \Delta \vdash B}\ \to\text{-E}$$

$$\frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \times B}\ \times\text{-I} \qquad \frac{\Gamma \vdash A \times B \qquad \Delta, A, B \vdash C}{\Gamma, \Delta \vdash C}\ \times\text{-E}$$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A + B}\ +\text{-I}_1 \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A + B}\ +\text{-I}_2$$

$$\frac{\Gamma, A \vdash C \qquad \Delta, A \vdash C \qquad \Delta, B \vdash C}{\Gamma, \Delta \vdash C}\ +\text{-E}$$

Figure 3.1: Multiple Proof Rules in Columns

3.2 Rules Intuitionistic Logic with Types

$$\frac{}{x : A \vdash x : A}\ \text{Id} \qquad \frac{\Gamma, \Delta \vdash t : A}{\Delta, \Gamma \vdash t : A}\ \text{Exchange}$$

$$\frac{\Gamma, y : A, z : A \vdash u : B}{\Gamma, x : A \vdash u[y \mapsto x][z \mapsto x] : B}\ \text{Contraction}$$

$$\frac{\Gamma \vdash u : B}{\Gamma, x : A \vdash u : B}\ \text{Weakening}$$

$$\frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \lambda x.t : A \to B}\ \to\text{-I}$$

$$\frac{\Gamma \vdash s : A \to B \qquad \Delta \vdash t : A}{\Gamma, \Delta \vdash s(t) : B}\ \to\text{-E}$$

$$\frac{\Gamma \vdash t : A \qquad \Delta \vdash u : B}{\Gamma, \Delta \vdash (t, u) : A \times B}\ \times\text{-I}$$

$$\frac{\Gamma \vdash s : A \times B \qquad \Delta, x : A, y : B \vdash v : C}{\Gamma, \Delta \vdash \text{case } s \text{ of } (x, y) \to v : C}\ \times\text{-E}$$

$$\frac{\Gamma \vdash t : A}{\Gamma \vdash \text{inl}(t) : A + B}\ +\text{-I}_1 \qquad \frac{\Gamma \vdash u : B}{\Gamma \vdash \text{inr}(u) : A + B}\ +\text{-I}_2$$

$$\frac{\Gamma \vdash s : A + B \qquad \Delta, x : A \vdash v : C \qquad \Delta, y : B \vdash w : C}{\Gamma, \Delta \vdash \text{case } s \text{ of } \text{inl}(x) \to v;\ \text{inr}(y) \to w : C}\ +\text{-E}$$

Figure 3.2: Multiple Proof Rules in Columns

3.3 Rules Linear Logic

$$\frac{}{\langle A\rangle \vdash A}\ \langle\text{Id}\rangle \qquad \frac{}{[A] \vdash A}\ [\text{Id}] \qquad \frac{\Gamma, \Delta \vdash A}{\Delta, \Gamma \vdash A}\ \text{Exchange}$$

$$\frac{\Gamma, [A], [A] \vdash B}{\Gamma, [A] \vdash B}\ \text{Contraction} \qquad \frac{\Gamma \vdash B}{\Gamma, [A] \vdash B}\ \text{Weakening}$$

$$\frac{[\Gamma] \vdash A}{[\Gamma] \vdash\ !A}\ !\text{-I} \qquad \frac{\Gamma \vdash\ !A \qquad \Delta, [A] \vdash B}{\Gamma, \Delta \vdash B}\ !\text{-E}$$

$$\frac{\Gamma, \langle A\rangle \vdash B}{\Gamma \vdash A \multimap B}\ \multimap\text{-I} \qquad \frac{\Gamma \vdash A \multimap B \qquad \Delta \vdash A}{\Gamma, \Delta \vdash B}\ \multimap\text{-E}$$

$$\frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \otimes B}\ \otimes\text{-I}$$

$$\frac{\Gamma \vdash A \otimes B \qquad \Delta, \langle A\rangle, \langle B\rangle \vdash C}{\Gamma, \Delta \vdash C}\ \otimes\text{-E}$$

$$\frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B}\ \&\text{-I}$$

$$\frac{\Gamma \vdash A \,\&\, B}{\Gamma \vdash A}\ \&\text{-E}_1 \qquad \frac{\Gamma \vdash A \,\&\, B}{\Gamma \vdash B}\ \&\text{-E}_2$$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \oplus B}\ \oplus\text{-I}_1 \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \oplus B}\ \oplus\text{-I}_2$$

$$\frac{\Gamma \vdash A \oplus B \qquad \Delta, \langle A\rangle \vdash C \qquad \Delta, \langle B\rangle \vdash C}{\Gamma, \Delta \vdash C}\ \oplus\text{-E}$$

Figure 3.3: Multiple Proof Rules in Columns

3.4 Rules Linear Logic with Types

$$\frac{}{\langle x : A\rangle \vdash x : A}\ \langle\text{Id}\rangle \qquad \frac{}{[x : A] \vdash x : A}\ [\text{Id}]$$

$$\frac{\Gamma, \Delta \vdash t : A}{\Delta, \Gamma \vdash t : A}\ \text{Exchange} \qquad \frac{\Gamma \vdash u : B}{\Gamma, [x : A] \vdash u : B}\ \text{Weakening}$$

$$\frac{\Gamma, [y : A], [z : A] \vdash u : B}{\Gamma, [x : A] \vdash u[y \mapsto x][z \mapsto x] : B}\ \text{Contraction}$$

$$\frac{[\Gamma] \vdash t : A}{[\Gamma] \vdash\ !t : !A}\ !\text{-I}$$

$$\frac{\Gamma \vdash s : !A \qquad \Delta, [x : A] \vdash u : B}{\Gamma, \Delta \vdash \text{case } s \text{ of } !x \to u : B}\ !\text{-E}$$

$$\frac{\Gamma, \langle x : A\rangle \vdash u : B}{\Gamma \vdash \lambda\langle x\rangle.u : A \multimap B}\ \multimap\text{-I}$$

$$\frac{\Gamma \vdash s : A \multimap B \qquad \Delta \vdash t : A}{\Gamma, \Delta \vdash s\langle t\rangle : B}\ \multimap\text{-E}$$

$$\frac{\Gamma \vdash t : A \qquad \Delta \vdash u : B}{\Gamma, \Delta \vdash \langle t, u\rangle : A \otimes B}\ \otimes\text{-I}$$

$$\frac{\Gamma \vdash s : A \otimes B \qquad \Delta, \langle x : A\rangle, \langle y : B\rangle \vdash v : C}{\Gamma, \Delta \vdash \text{case } s \text{ of } \langle x, y\rangle \to v : C}\ \otimes\text{-E}$$

$$\frac{\Gamma \vdash t : A \qquad \Gamma \vdash u : B}{\Gamma \vdash \langle\langle t, u\rangle\rangle : A\ \&\ B}\ \&\text{-I}$$

$$\frac{\Gamma \vdash s : A\ \&\ B}{\Gamma \vdash \text{fst}\langle s\rangle : A}\ \&\text{-E}_1 \qquad \frac{\Gamma \vdash s : A\ \&\ B}{\Gamma \vdash \text{snd}\langle s\rangle : B}\ \&\text{-E}_2$$

$$\frac{\Gamma \vdash t : A}{\Gamma \vdash \text{inl}\langle t\rangle : A \oplus B}\ \oplus\text{-I}_1 \qquad \frac{\Gamma \vdash u : B}{\Gamma \vdash \text{inr}\langle u\rangle : A \oplus B}\ \oplus\text{-I}_2$$

$$\frac{\Gamma \vdash s : A \oplus B \qquad \Delta, \langle x : A\rangle \vdash v : C \qquad \Delta, \langle y : B\rangle \vdash w : C}{\Gamma, \Delta \vdash \text{case } s \text{ of } \text{inl}\langle x\rangle \to v;\ \text{inr}\langle y\rangle \to w : C}\ \oplus\text{-E}$$

Figure 3.4: Multiple Proof Rules in Columns

Chapter 4

Call-by-box Lambda Calculus

4.1 Specification of λ_b

4.1.1 Intuitionistic logic IS4

Intuitionistic logic IS4 is a non-classical logic that is based on the idea of constructive proof. Unlike classical logic, intuitionistic logic does not allow for the law of excluded middle, which states that every proposition is either `true` or `false`. Intuitionistic logic also does not allow for the law of double negation, which states that if a proposition is not `false`, then it is `true`. Thus, if we know that proposition A not `false`, we cannot conclude that it is `true`. In intuitionistic logic, a proposition is only considered `true` if it can be constructively proven. That is, a proposition is `true` if we can build a proof step by step, much like writing a program. This is why programming languages and proofs in systems like Agda and Coq are based on intuitionistic logic. Since lambda calculus is the foundation of functional programming, it corresponds with intuitionistic logic via the Curry-Howard correspondence, where programs are equivalent to proofs and types to propositions.

The intuitionistic modal logic IS4 is an extension of intuitionistic logic that introduces *modal operators*. The $\lambda_\Box$ calculus is a fragment of IS4 that, besides the usual λ-terms, also uses box (box) and unbox (ε) operators and is strong enough to interpret both call-by-name (cbn) and call-by-value (cbv) lambda calculi. The box is like a little container that delays a computation, making it possible to simulate both cbn and cbv evaluation strategies. In this research, we formalize the *boxed lambda calculus* in Agda, denoted as λ_b, which is as a refinement of the simpler $\lambda_\Box$ calculus.

4.1.2 Terms and Types of λ_b

We now define the types of the λ_b calculus. We indicate the types of λ_b with A and B, representing types and boxed types respectively. The type X denotes the base type. Note that functions always take boxed types as input.

Definition 4.1 (Types of λ_b)**.**

$$A \quad ::= \quad X \mid \ B \to A \mid \ B \qquad\qquad B \quad ::= \quad \Box A$$

We give some examples of types in λ_b. Note that A can be any type that is not boxed, while B can be any type that is boxed. The type X is the base type.

Example 4.2 (Types in λ_b)**.**

$$\begin{aligned} &X \\ &\Box(B \to A) \\ &\Box X \to X \\ &\Box A \to \Box B \\ &\Box\Box\Box X \to \Box X \end{aligned}$$

We now define the terms of λ_b. The terms are defined by the grammar in 4.3.

Definition 4.3 (Terms in λ_b)**.**

$$M, N, P, Q \quad ::= \quad \varepsilon(x) \mid \ \lambda x.M \mid \ MN \mid \ \mathsf{box}(N)$$

We give some examples of terms that are well typed in λ_b.

Example 4.4 (Terms in λ_b)**.** Assuming $\ldots$, the following terms are well-typed in λ_b:

$$\begin{aligned} &\varepsilon(x) \\ &\lambda x.\varepsilon(x) \\ &(\lambda x.\varepsilon(x))\mathsf{box}(\varepsilon(y)) \\ &(\lambda x.\varepsilon(x))\mathsf{box}((\lambda y.\varepsilon(y))) \end{aligned}$$

Definition 4.5. The typing rules of λ_b are give by:

$$\frac{}{\Gamma, x : \Box A \vdash \varepsilon(x) : A}\ \mathrm{Id}_\Box$$

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash \lambda x.M : A \to B}\ \to\text{-I}$$

$$\frac{\Gamma \vdash M : A \to B \qquad \Delta \vdash N : A}{\Gamma, \Delta \vdash MN : B} \to\text{-E}$$

$$\frac{\Gamma \vdash M : A}{\Gamma \vdash \mathsf{box}(M) : \Box A} \Box\text{-I}$$

Since variables always occur inside ε, we need to redefine free and bound variables and substitution. For λ_b, we have:

Definition 4.6 (Free and Bound Variables)**.** The free and bound variables of a term M are defined by:

$$\begin{array}{llll} FV_b(\varepsilon(x)) & = \{\varepsilon(x)\} & BV(\varepsilon(x)) & = \emptyset \\ FV_b(\lambda x.M) & = FV(M)\backslash\{\varepsilon(x)\} & BV(\lambda x.M) & = BV(M) \cup \{\varepsilon(x)\} \\ FV_b(MN) & = FV(M) \cup FV(N) & BV(MN) & = BV(M) \cup BV(N) \end{array}$$

Definition 4.7 (Substitution in λ_b)**.** The substitution of a term Q for a variable $\varepsilon(x)$ in a term M is defined as follows:

$$\begin{array}{ll} \varepsilon(x)[\varepsilon(x) \mapsto Q] & = Q \\ \varepsilon(y)[\varepsilon(x) \mapsto Q] & = \varepsilon(y) \text{ if } (x \neq y) \\ (MN)[\varepsilon(x) \mapsto Q] & = M[x \mapsto Q]N[x \mapsto Q] \\ (\lambda x.M)[\varepsilon(x) \mapsto Q] & = \lambda x.M \\ (\lambda y.M)[\varepsilon(x) \mapsto Q] & = \lambda z.M[\varepsilon(y) \mapsto \varepsilon(z)][\varepsilon(x) \mapsto Q] \text{ if } (x \neq y) \end{array}$$

with z a variable defined by:

1. If $\varepsilon(x) \notin FV_b(N)$ or $\varepsilon(y) \notin FV_b(M)$ then $z = y$
2. Otherwise, z can be any variable such that $\varepsilon(z) \notin FV(N)$ or $\varepsilon(z) \notin FV(M)$

Proposition 4.8 (Preservation of well-typedness)**.**

$$\Gamma, x : A \vdash M : B \text{ and } \Gamma \vdash Q : A \implies \Gamma \vdash M[\varepsilon(x) \mapsto Q] : B$$

Proof. We prove by structural induction on the term M that this substitution preserves well-typedness. *? is well-defined ?*

1. $$\varepsilon(x)[\varepsilon(x) := Q] = Q$$

   Since Q is well-typed, and Q is the result of the substitution, we have that well-typedness is preserved.

2. $$\varepsilon(y)[\varepsilon(x) := Q] = \varepsilon(y) \quad \text{if } x \neq y$$

   We have that $x \neq y$, and the result of the substitution is $\varepsilon(y)$, which is well-typed.

3.
$$(MN)[\varepsilon(x) := Q] = (M[\varepsilon(x) := Q])\ (N[\varepsilon(x) := Q])$$

We have that MN is well-typed and thus M and N are well-typed. By the inductive hypothesis, we get that $M[\varepsilon(x) := Q]$ is well-typed, and $N[\varepsilon(x) := Q]$ is well-typed. We also have that application of two well-typed terms is well-typed, so we have that $(MN)[\varepsilon(x) := Q]$ is well-typed.

4.
$$(\lambda x.M)[\varepsilon(x) := Q] = \lambda x.M$$

We have that $\lambda x.M$ is well-typed and thus M is well-typed. Since $\varepsilon(x)$ is a bound variable in $\lambda x.M$, no substitution occurs and the result is the well-typed term $\lambda x.M$.

5.
$$(\lambda y.M)[\varepsilon(x) := Q] = \lambda z.(M[\varepsilon(y) := \varepsilon(z)][\varepsilon(x) := Q]) \quad \text{if } x \neq y$$

We have that $\lambda y.M$ is well-typed and thus M is well-typed. By the inductive hypothesis, we have that $M[\varepsilon(y) := \varepsilon(z)]$ and $M[\varepsilon(y) := \varepsilon(z)][\varepsilon(x) := Q]$ are well-typed. We also havee that abstraction of a well-typed term is well-typed, so we have that
$\lambda z.(M[\varepsilon(y) := \varepsilon(z)][\varepsilon(x) := Q])$ is well-typed.

□

The reduction rule we can use for this λ-calculus is:

Definition 4.9 (β_b)**.** In λ_b, β reduction is defined as follows:

$$(\lambda x.M)\mathsf{box}(N) \to M[\varepsilon(x) \mapsto N] \qquad (\beta_b)$$

We define three relations: β_b closed under μ and ν, β_b closed under μ and β_b closed under ν. We call them $\to_{we}$, $\to_{b_n}$ and $\to_{b_v}$ respectively. The relation $\to_{we}$ is weak since values do not reduced. That is, abstractions and terms of the form $\varepsilon(x)$ do not reduce. The relation $\to_{we}$ is also external as boxes do not reduce. This relation is called call-by-box (cbb) evaluation. We will see after defining the embeddings that $\to_{we}$ collapses to $\to_{b_n}$ and $\to_{b_v}$ when we restrict the terms to terms of the image of Girard's or Gödel's embedding.

Relation	$\boldsymbol{\beta}$	$\boldsymbol{\mu}$	$\boldsymbol{\mu_>}$	$\boldsymbol{\nu}$	$\boldsymbol{\nu_<}$	$\boldsymbol{\xi}$
$\rightarrow_{\beta_n}$	β_n	✓	-	✓	✓	✓
$\rightarrow_n$	β_n	✓	-			
$\rightarrow_w$	β_n	✓	-	✓		
$\rightarrow_{\beta_v}$	β_v	✓	-	✓	✓	✓
$\rightarrow_v$	β_v	✓	-		✓	
$\rightarrow_{\beta_b}$	β_b	✓	✓	✓	✓	✓
$\rightarrow_{we}$	β_b	✓	✓	✓		

Table 4.1: Closure Rules for Different Reduction Relations

4.2 Embedding of λ_n into λ_b

Let Λ_{CBN} be the set containing all λ-terms in the λ_n and let Λ_{CBB} be the set containing all λ-terms in the λ_b. The set $\mathcal{T}_{CBN}$ contains all types of the λ_n and $\mathcal{T}_{CBB}$ contains all types of the λ_b. Now we define functions from Λ_{CBN} to Λ_{CBB} for term translation and from $\mathcal{T}_{CBN}$ to $\mathcal{T}_{CBB}$ for type translation. These functions represent an embedding of the λ_n into the λ_b.

$$
\begin{array}{lll@{\qquad}lll}
T_t & : \mathcal{T}_{CBN} \rightarrow \mathcal{T}_{CBB} & & T & : \Lambda_{CBN} \rightarrow \Lambda_{CBB} \\
T_t(X) & = X & & T(x) & = \varepsilon(x) \\
T_t(A \rightarrow B) & = \Box T_t(A) \rightarrow T_t(B) & & T(\lambda x.M) & = \lambda x.T(M) \\
 & & & T(MN) & = T(M)\mathsf{box}(T(N))
\end{array}
$$

The image of Girard's translation is a subset of the λ_b givem by the following grammar:

$$M, N \quad ::= \quad \varepsilon(x) \mid \ \lambda x.M \mid \ M\mathsf{box}(N)$$

Since application in Girard's translation is of the form

$$M\mathsf{box}(N)$$

reducing the left-hand side of the application is not possible without the ζ rule that allows reduction inside boxed terms.

To clarify, we give the embedding of $\mathbf{I}x$ into λ_b after which we apply $\rightarrow_{b_n}$. It is clear that the result obtained after $\rightarrow_{b_n}$ is equal to the embedding of the result of applying $\rightarrow_{b_n}$ on $\mathbf{I}x$ in the λ_n, which is $T(x) = \varepsilon(x)$.

$$
\begin{array}{rcl}
T(\mathbf{I}x) & = & (T(\lambda x.x))\mathsf{box}(T(x)) \\
& = & (\lambda x.T(x))\mathsf{box}(T(x)) \\
& = & (\lambda x.\varepsilon(x))\mathsf{box}(\varepsilon(x)) \\
& \to_{b_n} & (\lambda x.\varepsilon(x))[\varepsilon(x) \mapsto \varepsilon(x)] \\
& = & \varepsilon(x)
\end{array}
$$

4.3 Embedding of λ_v into λ_b

For the embedding of the λ_v into the λ_b, we make use of the abbreviation "raise" defined by:

$$\text{raise}(M) := \lambda z.\varepsilon(z)M$$

Let Λ_{CBV} be the set containing all λ-terms in the λ_v and let Λ_{CBB} be the set containing all λ-terms in the λ_b. The set $\mathcal{T}_{CBV}$ contains all types of the λ_v and $\mathcal{T}_{CBB}$ contains all types of the λ_b. Now we define functions from Λ_{CBV} to Λ_{CBB} for term translation and from $\mathcal{T}_{CBV}$ to $\mathcal{T}_{CBB}$ for type translation.

$$
\begin{array}{lllll}
T_t & : \mathcal{T}_{CBV} \to \mathcal{T}_{CBB} & \qquad & T & : \Lambda_{CBN} \to \Lambda_{CBB} \\
T_t(X) & = \Box X & & T(x) & = \mathsf{box}(\varepsilon(x)) \\
T_t(A \to B) & = \Box(\Box T_t(A) \to \Box T_t(B)) & & T(\lambda x.M) & = \mathsf{box}(\lambda x.T(M)) \\
& & & T(MN) & = \text{raise}(T(N))T(M)
\end{array}
$$

To clarify, we give the embedding of $\mathbf{I}x$ into λ_b after which we apply $\to_{b_v}$ twice. It is clear that the result obtained after $\to_{b_v}$ is equal to the embedding of the result of applying $\to_v$ on $\mathbf{I}x$ in the λ_v, which is $T(x) = \mathsf{box}(\varepsilon(x))$.

$$
\begin{array}{rcl}
T(\mathbf{I}x) & = & \text{raise}(T(x))T(\mathbf{I}) \\
& = & \text{raise}(T(x))T(\lambda x.x) \\
& = & \text{raise}(\mathsf{box}(\varepsilon(x)))\mathsf{box}(\lambda x.T(x)) \\
& = & \text{raise}(\mathsf{box}(\varepsilon(x)))\mathsf{box}(\lambda x.\mathsf{box}(\varepsilon(x))) \\
& = & (\lambda z.\varepsilon(z)\mathsf{box}(\varepsilon(x)))\mathsf{box}(\lambda x.\mathsf{box}(\varepsilon(x))) \\
& \to_{b_v} & (\lambda x.\mathsf{box}(\varepsilon(x)))\mathsf{box}(\varepsilon(x)) \\
& \to_{b_v} & \mathsf{box}(\varepsilon(x))
\end{array}
$$

Chapter 5

Formalisation in Agda

5.1 Box Calculus

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5.2 Linear Logic

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec tincidunt, nunc ut bibendum facilisis, nisi nunc fringilla nisi, nec mollis nunc nisi nec ligula. Donec id ligula ac enim efficitur mollis. Donec id ligula ac enim efficitur mollis. Donec id ligula ac enim efficitur mollis. Donec id ligula ac enim efficitur mollis.

Chapter 6

Related Work

In this chapter you demonstrate that you are sufficiently aware of the state-of-art knowledge of the problem domain that you have investigated as well as demonstrating that you have found a *new* solution / approach / method.

Chapter 7

Conclusions

In this chapter you present all conclusions that can be drawn from the preceding chapters. It should not introduce new experiments, theories, investigations, etc.: these should have been written down earlier in the thesis. Therefore, conclusions can be brief and to the point.

Bibliography

[1] Simon Peyton Jones. How to write a good research paper, 2004. Presentation at Technical University of Vienna, http://research.microsoft.com/en-us/um/people/simonpj/papers/giving-a-talk/writing-a-paper-slides.pdf.

Appendix A

Appendix

Appendices are *optional* chapters in which you cover additional material that is required to support your hypothesis, experiments, measurements, conclusions, etc. that would otherwise clutter the presentation of your research.